

Robbie – “*Object Detected and Moved*”

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Abstract

This project presents Robbie, a **3-DOF** robotic arm designed to autonomously detect and manipulate objects using low-cost, off-the-shelf components. The system combines computer vision (**YOLOv8** + **OpenCV**), embedded motion control (**Arduino Nano**), and visual input from an **ESP32-CAM**. By translating object detection results into inverse kinematics (IK) commands, Robbie demonstrates how a cost-effective robotic manipulator can be constructed for educational and research purposes.

Introduction

Industrial robotic manipulators are typically expensive and rely on high-precision actuators and sensors, placing them out of reach for many students and hobbyists. Robbie addresses this gap by providing a DIY, low-cost alternative that can be replicated with minimal resources.

Inspired by fictional assistants such as *DUM-E* from *Ironman* (Yes, I am a fan), Robbie was conceived as a hands-on introduction to robotics, machine vision, and kinematic modeling. The initial prototype was constructed from cardboard and broomsticks, later upgraded with 3D-printed parts for precision and durability.

Key Features:

- Uses widely available microcontrollers (Arduino Nano, ESP32-CAM).
- Custom-trained YOLOv8 object detection model.
- Fully documented workflow for reproducibility.
- Serves as a foundation for more advanced autonomous systems.
- Innovative 3D designed alternatives inspired from tower cranes and LEGO.

System Architecture

Input

- ESP32-CAM captures still images of the workspace.
- Python script requests images via HTTP for processing.

Processing

- YOLOv8 detects objects and outputs bounding box data.
- OpenCV overlays a reference grid and maps object centers.
- Coordinates are transmitted to the Arduino Nano via serial communication.

Output

- Polar coordinates (r, θ) drive the base stepper rotation and link servos.
- Inverse kinematics compute link angles for target positioning.
- Gripper engages, transfers the object to a drop zone, and releases it.
- Arduino signals completion to the pc as “**DONE**” to start the next detection cycle.

Hardware Design

Actuators

- 28BYJ-48 Stepper Motor – base rotation.
- Two MG996R Servos – control primary arm links.
- SG90 Servo – shovel-style gripper.

Sensor

- ESP32-CAM (external antenna) – workspace image acquisition.

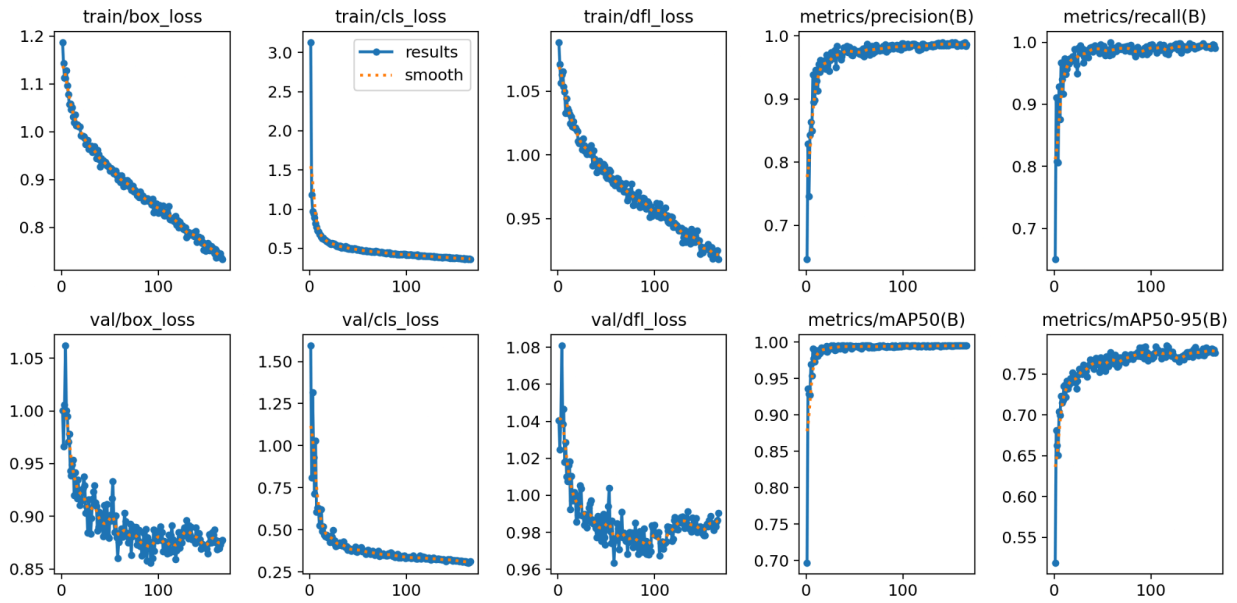
Materials

- Initial prototype: broomsticks + layered cardboard.
- Final version: 3D-printed components for strength and precision.

Software Design

Object Detection Training

- Model: YOLOv8s, custom-trained.
- Dataset: ~100 images per object class, orientations (top, side, bottom).
- Environment: Controlled lighting, white A4 background.
- Training: ~190 epochs, achieving:
 - $\text{mAP}@50 = 0.995$
 - $\text{mAP}@50-95 \approx 0.78$
 - Precision & Recall ≈ 0.99

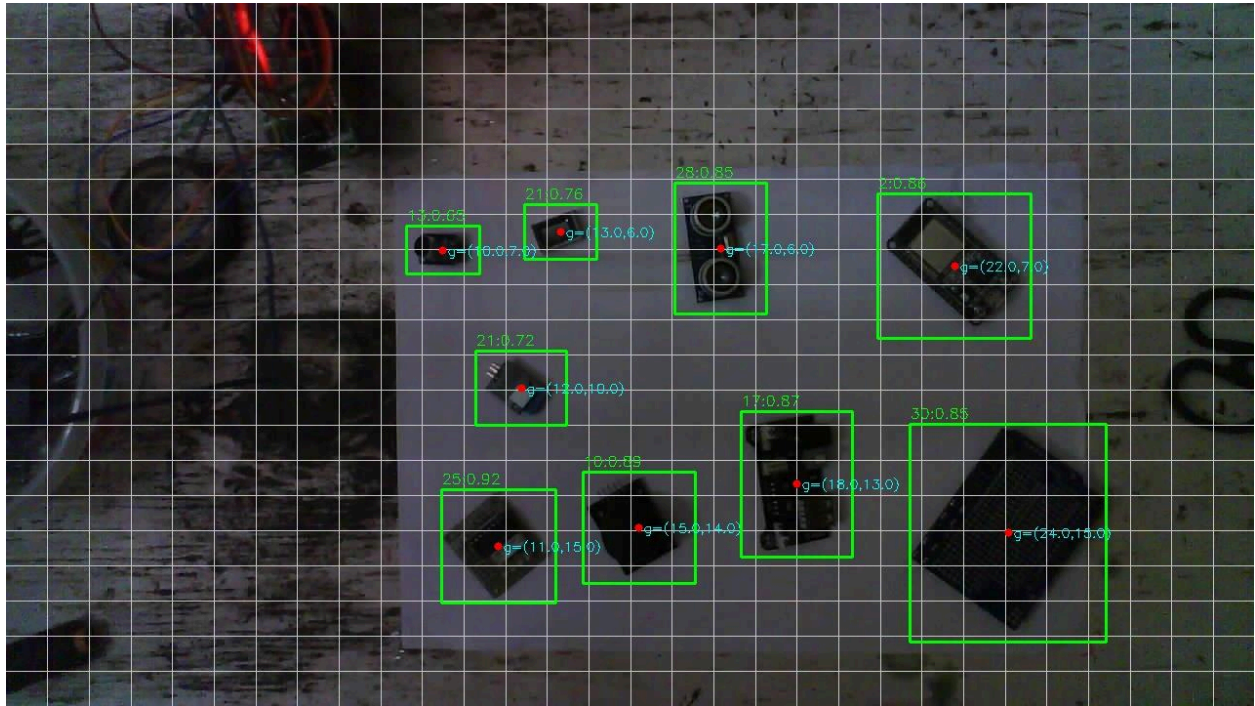


Despite very high accuracy, minor false positives are acceptable; the manipulator can still attempt pickups successfully.

Python + OpenCV Workflow

- Requests image from ESP32-CAM.
- YOLO processes image → bounding box details saved.
- Object centers mapped to grid coordinates.
- Real-world XY positions sent to Arduino.
- Annotated images optionally saved for visualization.

Detected Windows Object



Arduino Actuation

1. Receive Coordinates: XY positions in cm.
2. Translation & Scaling: Apply workspace offsets (transX, transY).
3. Polar Conversion: Convert $XY \rightarrow (r, \theta)$.
4. Inverse Kinematics: Compute joint angles for target reach.
5. Execution:
 - Stepper rotates by θ .
 - Servos move smoothly to target via `smoothMove()`.
 - Gripper closes, transports, and releases object.
6. Loop: Arduino signals **DONE** and waits for input from master script.

Geometry and Kinematics

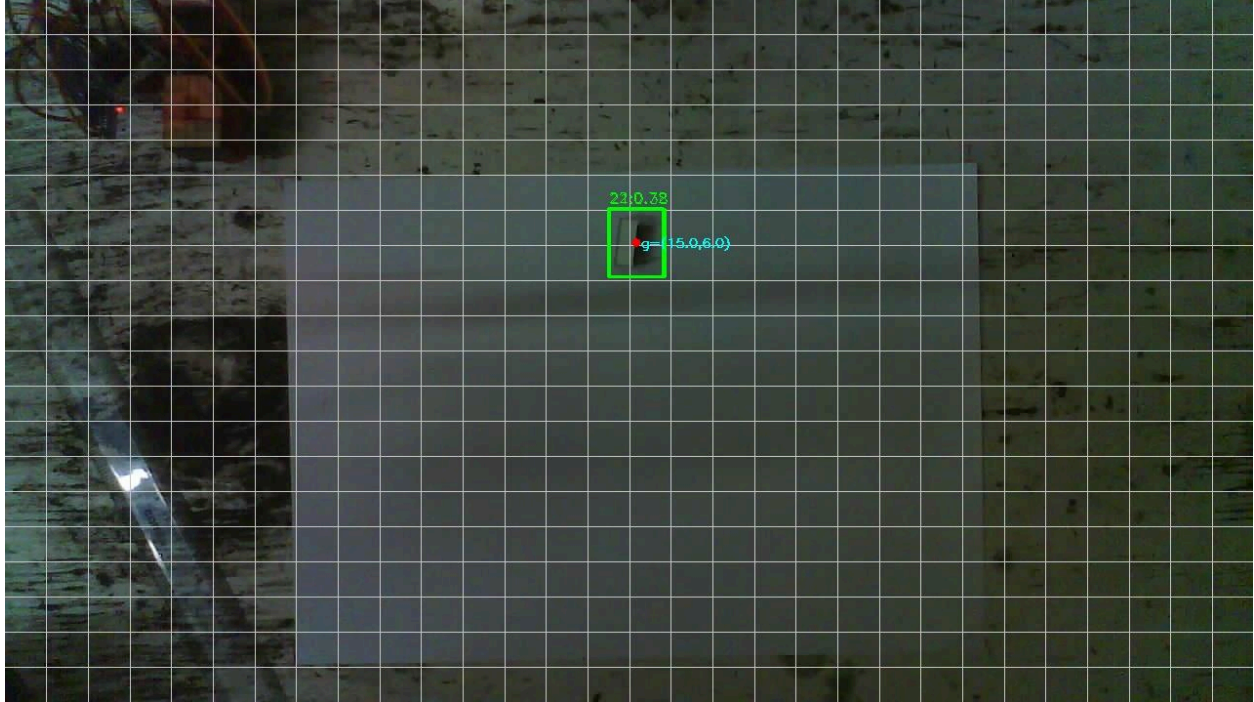
Key Dimensions

- Base stepper to origin: 3 cm.
- Stepper to workspace edge: ~ 3 cm.
- Bottom servo height: ~ 3.5 cm.
- Link 1 (LB): 16 cm.
- Link 2 (LT): 10.5 cm.
- Gripper clearance (GC): 4 cm.

Kinematic Model

Robbie is modeled as a 2-link planar manipulator with base offset.

- Input coordinates (pixels) $\rightarrow$ scaled to real-world XY.
- Arduino converts XY $\rightarrow$ polar (r, θ).
- Inverse kinematics solved using cosine law to compute joint angles.
- Offsets for clearances (SC, GC, LB, LT) included in calculations.



Software Flow Diagram

Python Script

1. Acquire image.
2. Detect objects (YOLOv8).
3. Map bounding boxes $\rightarrow$ grid $\rightarrow$ XY coordinates.
4. Send XY via serial.

Arduino Code

1. Read XY coordinates.
2. Convert to polar + apply IK.
3. Execute motor and servo motions.
4. Operate gripper.

5. Return to the waiting position.

Constraints and Safety

- Valid workspace: $7\text{ cm} < r < 22.5\text{ cm}$.
- Base rotation limited to -90° to $+180^\circ$.
- Motions aborted if inputs exceed limits.
- Servos actuated sequentially to avoid overshoot.
- Stepper operated on an open-loop with step count tracking.

AI Model Validation

Ground Truth Annotations (Validation Set):



Predictions (YOLOv8 Model):



Robbie's trained model performs robustly under real-world lighting variations and clutter. Occasional misclassifications are tolerated since the manipulator operates with redundancy (attempts are still valid picks).

Video

 ROBBIE_LIVES.mp4

Conclusion

Robbie demonstrates that computer vision-driven robotic manipulation can be achieved with accessible tools and modest hardware. This project merges fundamental EE concepts: control systems, embedded programming, kinematics, and machine learning.

It stands as a practical entry point for students worldwide into robotics research, blending low-cost hardware with cutting-edge AI in an academic framework suitable for undergraduate engineering applications.